RAHUL KUMAR

STAFF SITE RELIABILITY ENGINEER | CLOUD INFRASTRUCTURE & DISTRIBUTED SYSTEMS

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EXECUTIVE SUMMARY

Staff Site Reliability Engineer with deep expertise in architecting, automating, and operating high-throughput multi-cloud distributed platforms (AWS, GCP, Azure) and event streaming architectures (Apache Kafka). Specialized in proactive reliability engineering—defining org-wide SLO/SLA frameworks, error budget policies, and AI-augmented incident response ecosystems. Trusted Incident Commander for Tier-0 critical outages with extensive experience authoring Customer Root Cause Analysis (CRCA) briefs, facilitating blameless post-mortems, and scaling Kubernetes-native platforms processing millions of events per second with 99.99% availability.

CORE COMPETENCIES & TECHNICAL PROFICIENCY

Distributed Systems & Kafka:	Apache Kafka (Brokers, Partitioning, ISR, KRaft Consensus, High Watermark vs LEO, Tiered Storage, Exactly-Once Semantics, Consumer Lag, Cooperative Sticky Assignors), Event-Driven Architecture.
Proactive Reliability & SRE:	SLO/SLA Architecture, Error Budget Policies, Incident Commander (IC), Blameless Post-Mortems, Customer Root Cause Analysis (CRCA), FMEA, Chaos Engineering, Capacity Planning.
Incident & Automation Tooling:	Rootly (Workflows & Slack Integrations), PagerDuty, Jira, Confluence, Slack Enterprise Grid Bots, AI-Assisted Runbooks, Automated Timeline & Post-Mortem Synthesis.
Cloud, Kubernetes & OS:	Multi-Cloud (AWS, GCP, Azure), Kubernetes (EKS/GKE), Istio Service Mesh, StatefulSets, Karpenter Autoscaling, Pod Anti-Affinity, Linux Kernel (Zero-Copy `sendfile()`, Page Cache, TCP Sockets).
Observability, IaC & CI/CD:	OpenTelemetry (Distributed Tracing), Prometheus, Grafana, Datadog, ELK Stack, Terraform, Terragrunt, Helm, ArgoCD (GitOps), GitHub Actions, Python, Go, Bash Scripting.

PROFESSIONAL EXPERIENCE

Lead Site Reliability & Cloud Platform Engineer — Programming.com (Client: Allcargo Gati) March 2024 – Present | New Delhi / Remote

Multi-account AWS architecture (PROD, SIT, DEV, UAT) supporting iTopaz Logistics, iTariff, and Prefiler streaming 50M+ events daily across Frankfurt (eu-central-1) and Singapore.

- **Proactive Reliability & SLO Governance:** Architected org-wide Service Level Objective (SLO) and Error Budget frameworks across 40+ microservices (`itariff-api`, `itopaz-platform`, `prefiler`); leveraged error budget burn-rate alerts to govern release velocity, reducing high-severity production incidents by **48%** within 12 months.
- **Rootly & PagerDuty Incident Modernization:** Spearheaded overhaul of incident response; integrated Rootly workflows with PagerDuty, Slack, and Jira, automating channel provisioning, role assignments (IC, Scribe, Comms Lead), and MTTR tracking. Slashed MTTA from **14 mins to < 90 seconds**.
- **Incident Command & Customer RCA (CRCA):** Served as primary Escalation Incident Commander for Tier-0/1 production outages during peak shipping events. Authored and peer-reviewed **45+ Customer Root Cause Analysis (CRCA)** briefs for enterprise clients with 100% adherence to 48-hour SLAs.
- **Distributed Streaming & Kafka Optimization:** Scaled multi-tenant Kafka clusters and Aurora PostgreSQL pipelines. Optimized Linux OS page cache writeback ratios (`dirty_ratio`), tuned broker TCP socket buffers (`net.core.rmem_max`), and resolved rebalance storms via Cooperative Sticky Assignors, mitigating consumer lag by **72%**.
- **Multi-Region EKS & Istio Mesh Resilience:** Designed multi-region AWS EKS clusters (`euc1-itopaz-prd-eks`, `apse1-itopaz-sit-eks`) with Karpenter autoscaling, strict pod anti-affinity, and Istio Service Mesh (mTLS, virtual services). Deployed GitOps zero-downtime pipelines via ArgoCD, sustaining **99.99% availability**.
- **Automation, FinOps & AI Incident Intelligence:** Built EventBridge/Lambda serverless schedulers (`scheduler-rds-ec2`) optimizing cluster spend, and developed an internal GenAI pipeline ingesting Slack transcripts and PagerDuty telemetry to auto-draft post-mortem timelines, saving ~4 hours per incident.

Founding Site Reliability & Platform Engineer — VCorp Tech (Incubated at DTU-IIF) 2023 – 2025 | Delhi, India

High-availability distributed cloud systems and AI application infrastructure incubated at DTU Innovation & Incubation Foundation.

- **Distributed Systems Architecture:** Architected containerized cloud infrastructure on Kubernetes and AWS, establishing multi-AZ microservice deployment topologies and automated GitOps CI/CD pipelines.
- **SRE & Incident Readiness:** Implemented Prometheus, Grafana, and OpenTelemetry observability stacks with automated alerting; authored standard operating procedures (SOPs) and runbooks for zero-downtime releases.
- **Data Streaming & Caching:** Designed asynchronous event pipelines utilizing Kafka/Redis for real-time telemetry processing, reducing message delivery latency by **40%**.

Senior Site Reliability & Cloud Infrastructure Engineer — Carrier Midea India (Midea) October 2020 – December 2022 | Gurugram / New Delhi

Cloud platform engineering, automation, and reliability leadership for enterprise commercial operations and IoT streaming services.

- **Multi-Cloud Infrastructure as Code:** Architected modular Terraform configurations to provision multi-AZ AWS environments (VPC, EKS, RDS Aurora, SQS, CloudWatch) complying with enterprise CIS benchmarks.
- **Incident Response & Runbook Automation:** Acted as on-call incident lead for platform stability; built automated PagerDuty escalation policies, self-healing health checks, and runbooks that reduced MTTR by **38%**.
- **Observability & Distributed Monitoring:** Deployed centralized Prometheus, Grafana, and ELK clusters; engineered dashboards tracking Golden Signals (Latency, Traffic, Errors, Saturation), eliminating false alarms by **55%**.
- **Resilience & Chaos Engineering:** Executed regular chaos experiments and game days validating auto-failover across multi-AZ RDS clusters and event broker replicas during simulated network partitions.

EDUCATION & ACADEMIC CREDENTIALS

Master of Business Administration (MBA) – Business Analytics
Delhi Technological University (DTU, formerly DCE) | *2023 – 2025*
Focus: Quantitative Systems Modeling, Large-Scale Analytics, Technology Strategy

Bachelor of Technology (B.Tech) – Information Technology
Delhi Technological University (DTU, formerly DCE) | *First Class with Distinction*
Focus: Operating Systems, Distributed Systems, Computer Networks, Database Systems

CERTIFICATIONS & CREDENTIALS

AWS Certified DevOps Engineer – Professional (DOP-C02) | **AWS Certified Data Engineer – Associate (DEA-C01)** | **AWS Knowledge: AI-Driven Development Lifecycle (97%)** | **AWS Knowledge: Cloud Essentials (96%)** | **AWS Cloud Quest: Cloud Practitioner**